

AI-Driven Multidisciplinary Design Optimization

AIAA MDO TC Workshops on Multifidelity Methods

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Siemens Digital Industries Software

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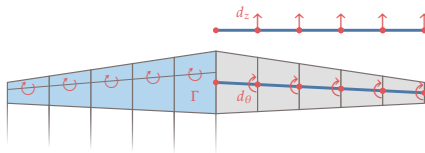
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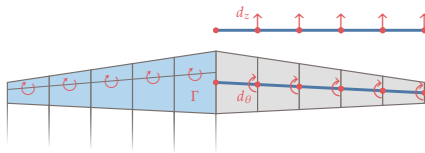
Multidisciplinary Design Optimization (1/2)

In the design of a wing, the design variables $\mathbf{x}$ are the airfoil, length, etc.



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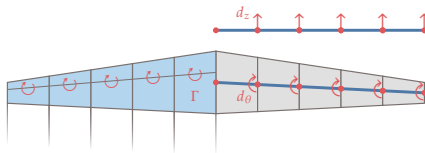


The designer wishes to minimize drag D subject to a stress constraint σ .

$$\begin{aligned} \min_{\mathbf{x}} \quad & D(\mathbf{x}; \hat{\Gamma}) \\ \text{subject to} \quad & \sigma(\mathbf{x}; \hat{\mathbf{d}}) - \sigma_{\max} \leq 0 \end{aligned}$$

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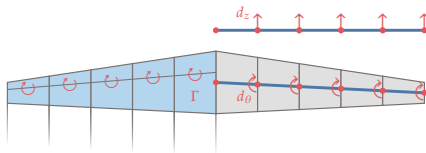
$$\text{subject to} \quad \sigma(\mathbf{x}; \hat{\mathbf{d}}) - \sigma_{\max} \leq 0$$

$$\text{while solving} \quad r_1(\Gamma; \hat{\mathbf{d}}, \mathbf{x}) = A(\hat{\mathbf{d}})\Gamma - \mathbf{v}(\hat{\mathbf{d}}) = \mathbf{0}$$

$$r_2(\mathbf{d}; \hat{\Gamma}, \mathbf{x}) = K\mathbf{d} - \mathbf{q}(\hat{\Gamma}) = \mathbf{0}$$

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The objective and constraint functions depend on two parameters $\hat{\Gamma}$ and $\hat{\mathbf{d}}$ which are computed using disciplinary analyses r_1 and r_2 .

Multidisciplinary Design Optimization (2/2)

Γ and $\mathbf{d}$ represent the aerodynamic vortices and structural displacements, respectively.

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$r_1(\Gamma; \hat{\mathbf{d}}) = \mathbf{0}$ and $r_2(\mathbf{d}; \hat{\Gamma}) = \mathbf{0}$ could involve computational fluid dynamics (CFD) and finite element analysis (FEA) softwares, respectively.

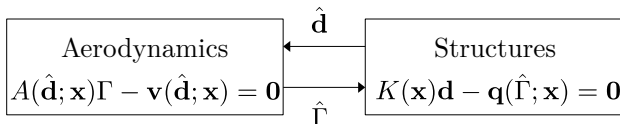
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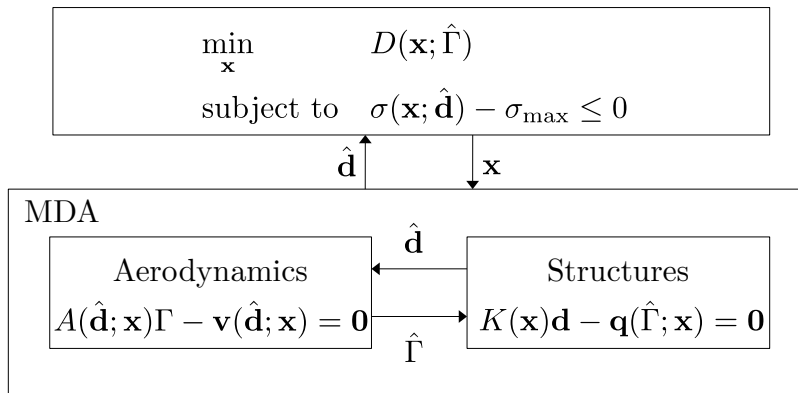
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A multidisciplinary analysis (MDA) is used to find $\hat{\Gamma}$ and $\hat{\mathbf{d}}$ that satisfy both r_1 and r_2 for a given wing design $\mathbf{x}$. $\hat{\Gamma}$ and $\hat{\mathbf{d}}$ are generally defined as *coupling variables*.



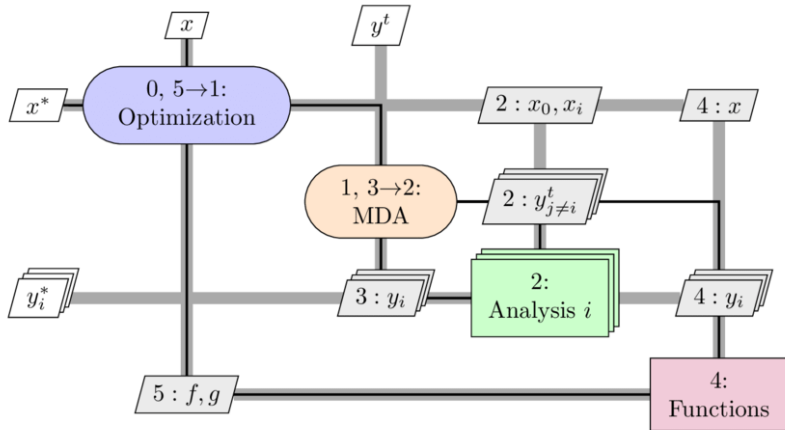
Multidisciplinary feasible (MDF) optimization (1/2)

We treat the converged MDA as a blackbox that simply returns the coupling variables as a function of the design variables $\hat{\mathbf{d}}(\mathbf{x})$ and attempt to perform optimization using an algorithm of our choice.



Block diagram of MDF

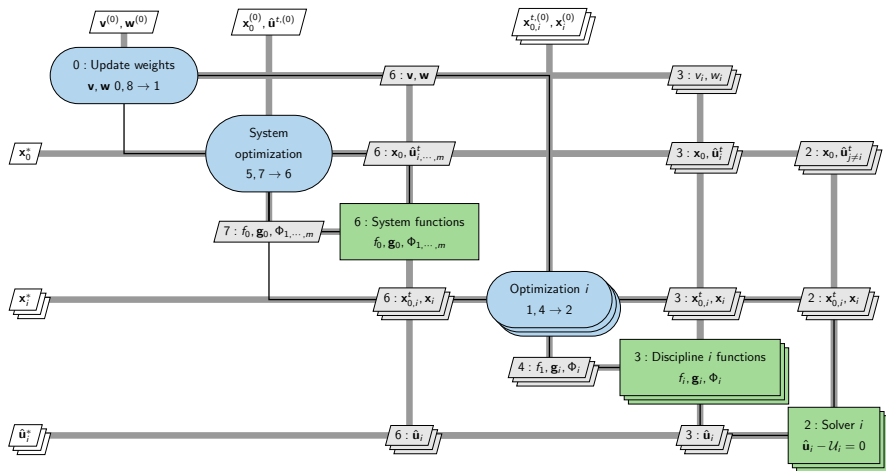
Multidisciplinary feasible (MDF) optimization (2/2)



Block diagram of MDF

Analytical target cascading (ATC)

We can represent the ATC formulation using the following XDSM.



XDSM of ATC formulation

Non-hierarchical analytical target cascading (NHATC) (1/3)

- We follow the same exact procedure as with ATC but instead of assigning objectives and constraints to system and subsystem problems we only do so for subsystem problems.

Non-hierarchical analytical target cascading (NHATC) (1/3)

- We follow the same exact procedure as with ATC but instead of assigning objectives and constraints to system and subsystem problems we only do so for subsystem problems.
- The general *subproblem* i (usually one for each $i \in \{1, \dots, m\}$):

$$\begin{aligned}
 &\underset{\mathbf{x}_{0,i}^t, \mathbf{x}_i, \hat{\mathbf{u}}_{j \neq i,i}^t}{\text{minimize}} && f_i(\mathbf{x}_{0,i}^t, \mathbf{x}_i, \hat{\mathbf{u}}_{j \neq i,i}^t; \hat{\mathbf{u}}_i) + \Phi_i(\hat{\mathbf{u}}_{j \neq i,i}^t - \hat{\mathbf{u}}_{j \neq i}) + \\
 &&& \sum_{j=1, j \neq i}^m \Phi_i(\mathbf{x}_{0,i}^t - \mathbf{x}_{0,j \neq i}^t, \hat{\mathbf{u}}_{i,j}^t - \hat{\mathbf{u}}_i) \\
 &\text{subject to} && \mathbf{g}_i(\mathbf{x}_{0,i}^t, \mathbf{x}_i, \hat{\mathbf{u}}_{j \neq i,i}^t; \hat{\mathbf{u}}_i) \leq \mathbf{0} \\
 &\text{while solving} && r_i(\hat{\mathbf{u}}_i; \mathbf{x}_{0,i}^t, \mathbf{x}_i, \hat{\mathbf{u}}_{j \neq i,i}^t) = \mathbf{0}, \text{ for } \hat{\mathbf{u}}_i
 \end{aligned}$$

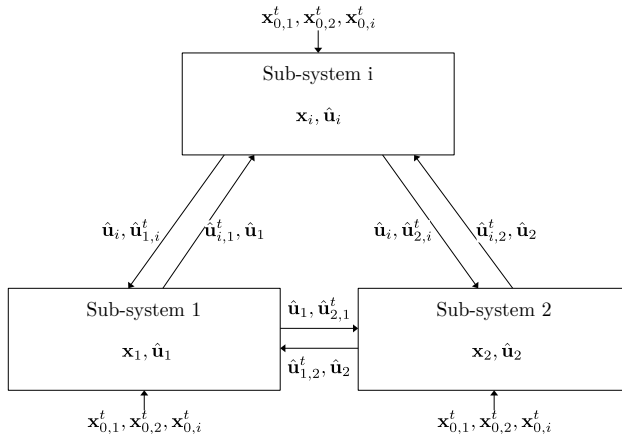
Non-hierarchical analytical target cascading (NHATC) (2/3)

This means that for subproblem $i \in \{1, \dots, m\}$:

- we formulate $j = m - 1$ consistency constraints between *non-local target coupling variables* $\hat{\mathbf{u}}_{i,j}^t$ and *local coupling variables* $\hat{\mathbf{u}}_i$ as $\mathbf{h}_j^c = \hat{\mathbf{u}}_{i,j}^t - \hat{\mathbf{u}}_i = \mathbf{0}$
- we formulate a consistency constraint between *local target coupling variables* $\hat{\mathbf{u}}_{j \neq i,i}^t$ and *non-local target coupling variables* $\hat{\mathbf{u}}_{j \neq i}$ as $\mathbf{h}_i^c = \hat{\mathbf{u}}_{j \neq i,i}^t - \hat{\mathbf{u}}_{j \neq i} = \mathbf{0}$
- we formulate $j = m - 1$ consistency constraints between *local target shared variables* $\mathbf{x}_{0,i}^t$ and *non-local target shared variables* $\mathbf{x}_{0,i}^t$ as $\mathbf{h}_{0,j}^c = \mathbf{x}_{0,i}^t - \mathbf{x}_{0,j}^t = \mathbf{0}$
- All subproblems are solved iteratively while updating penalty weights using ADMM coordination.

NHATC: Block diagram (3/3)

The previous NHATC formulation is shown in the following block diagram.

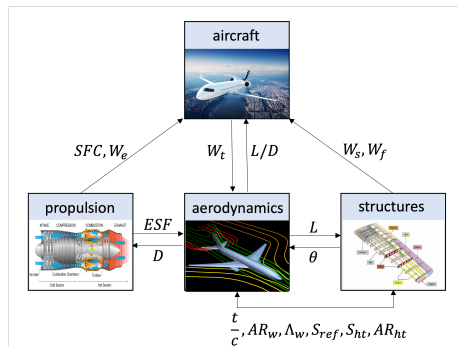


Block diagram of ATC formulation

MDO process Challenges

Key characteristics:

- Coupled physics / simulation models
- Coordination of multiple subproblems
- Large design spaces
- Conflicting objectives & constraints
- High computational cost



Blackbox Optimization Challenges

We consider blackbox optimization problems:

$$\min_{x \in \Omega} f(x)$$

where evaluations of f and the functions defining Ω are usually the result of a computer code (a blackbox).



BBO challenges



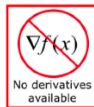
Long runtime



Large memory requirement



Software might fail



No derivatives available



Local optima



Non-smooth, noisy

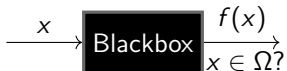
Possible bad properties of a blackbox from *J. Simonis, 2009, The Boeing Company*

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BBO challenges

The AI Opportunity

Machine learning can dramatically reduce evaluation cost, enable global search, and learn cross-disciplinary couplings from data.

AI use in Classical MDO Architectures

Architecture	Coupling strategy	Typical use	AI use
MDF	Fixed-point iteration	Tightly coupled systems	Surrogate modeling
IDF	Consistency constraints	Parallel disciplines	Bayesian optimization
AAO	All-at-once	Gradient-rich problems	Differentiable surrogates
CSSO	Coordination	Large organizations	RNNs and GRUs
BLISS	Sensitivity-based	Hierarchical systems	Sensitivity learning

Table: Common MDO architectures, their trade-offs and AI use as surrogate models.

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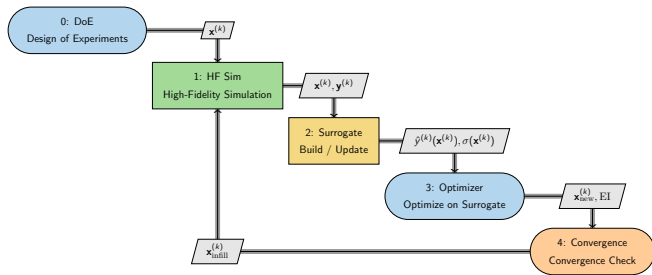
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Surrogate-Based Optimization (SBO) Workflow



Key steps: Sample $\rightarrow$ Simulate $\rightarrow$ Surrogate $\rightarrow$ Optimize $\rightarrow$ Check convergence

- SBO is a powerful framework for expensive MDO problems.
- Surrogate quality and infill strategy are critical.
- Key metric: **Expected Improvement (EI)** or **Probability of Feasibility**
- Infill criterion balances *exploitation* and *exploration*

Gaussian Process Surrogate — Quick Recap

A GP defines a distribution over functions:

$$f(\mathbf{x}) \sim \mathcal{GP}(\mu(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$$

Prediction

$$\begin{aligned}\mu^* &= \mathbf{k}_*^\top \mathbf{K}^{-1} \mathbf{y} \\ \sigma^{*2} &= k_{**} - \mathbf{k}_*^\top \mathbf{K}^{-1} \mathbf{k}_*\end{aligned}$$

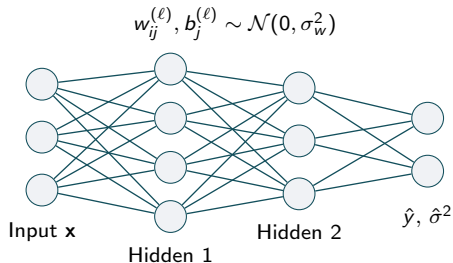
Common kernels

- Squared Exponential (RBF)
- Matérn 5/2
- Rational Quadratic

Key insight

σ^* gives a **free uncertainty estimate** — essential for active learning in MDO.

Bayesian Neural Network (BNN): Architecture and Probabilistic Prediction



$$p(y | \mathbf{x}, \mathcal{D}) = \int p(y | \mathbf{x}, \boldsymbol{\theta}) p(\boldsymbol{\theta} | \mathcal{D}) d\boldsymbol{\theta}$$

$$p(y | \mathbf{x}, \mathcal{D}) \approx \frac{1}{S} \sum_{s=1}^S p(y | \mathbf{x}, \boldsymbol{\theta}^{(s)}), \quad \boldsymbol{\theta}^{(s)} \sim p(\boldsymbol{\theta})$$

$$P(y \leq y_{\text{req}} | \mathbf{x}, \mathcal{D}) \approx \frac{1}{S} \sum_{s=1}^S \mathbb{I}(y^{(s)} \leq y_{\text{req}})$$

Quality/criteria probability

For pass/fail criteria (e.g., stress $\leq \sigma_{\text{max}}$), BNN outputs a full predictive distribution, enabling direct risk-aware decisions.

Physics-Informed Neural Networks (PINNs)

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{data}}}_{\text{observations}} + \lambda \underbrace{\mathcal{L}_{\text{PDE}}}_{\text{residual}} + \mu \underbrace{\mathcal{L}_{\text{BC}}}_{\text{boundaries}}$$

- Embed governing equations directly in the loss function
- Works with sparse / noisy data
- Suitable for **inverse problems** & sensitivity analysis
- Libraries: DeepXDE, Modulus (NVIDIA), JAX

MDO advantage

PINNs can serve as differentiable discipline models, enabling gradient propagation across the entire coupled system.

Agentic AI in MDO (1/4)

What is an agentic workflow?

- LLM-based planner decomposes design goals into tasks
- Tool-using agents call solvers, scripts, and databases
- Memory stores prior runs, constraints, and failures
- Verifier agent checks feasibility before promotion

Closed-loop cycle

Plan → Simulate → Learn → Propose → Verify

- Human sets objectives, safety limits, and trust gates
- Agent proposes candidates and justifies trade-offs

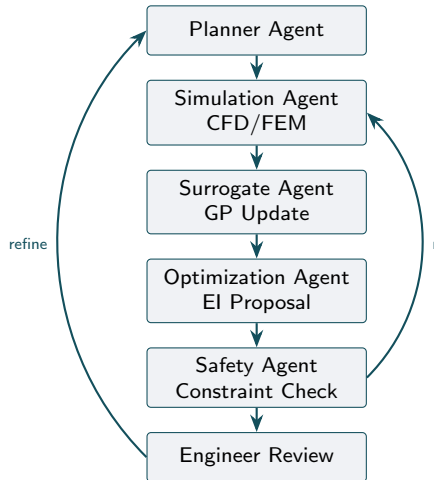
Why it matters for MDO

Agentic AI can orchestrate multi-disciplinary pipelines, reduce manual iteration overhead, and accelerate convergence while keeping engineers in the loop.

Agentic AI in MDO (2/4): Illustrative Example

Scenario: Wing drag minimization with lift and stress constraints

- 1 **Planner agent** decomposes goals and constraints
- 2 **Simulation agent** runs CFD/FEM on candidates
- 3 **Surrogate agent** updates GP mean and uncertainty
- 4 **Optimization agent** selects EI-driven infill points
- 5 **Safety agent** filters infeasible proposals
- 6 **Reviewer agent** sends ranked options to engineer



Related references: Yao et al., *ReAct* (ICLR 2023); Wu et al., *AutoGen* (2023); Martins & Ning, *Engineering Design*

Agentic AI in MDO (3/4): Illustrative Example - Wing MDO

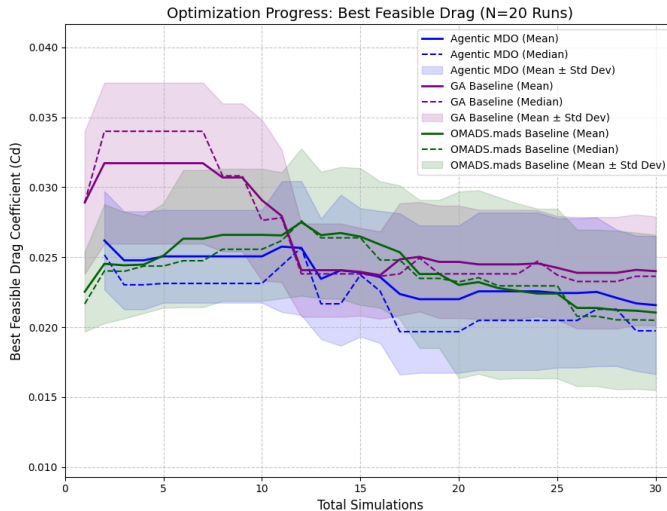
$$\begin{aligned}
 \min_{\mathbf{x}} \quad & f(\mathbf{x}) = C_d(\mathbf{x}) \\
 \text{s.t.} \quad & g_1(\mathbf{x}) = |C_l(\mathbf{x}) - 1.2| - 0.1 \leq 0, \\
 & g_2(\mathbf{x}) = \sigma(\mathbf{x}) - 800 \leq 0, \\
 & \mathbf{x}^L \leq \mathbf{x} \leq \mathbf{x}^U
 \end{aligned}$$

$$\mathbf{x} = [\text{AR}, \text{TR}, \text{tc}]^\top, \quad \mathbf{x}^L = [5.0, 0.2, 0.08]^\top, \quad \mathbf{x}^U = [15.0, 0.8, 0.15]^\top$$

$$C_d(\mathbf{x}) = 0.02 + \frac{0.01}{\text{AR}} + 0.05 \text{tc}^2 + 0.01(1 - \text{TR})$$

$$C_l(\mathbf{x}) = 0.5 + 0.05 \text{AR} + 0.1 \text{TR} - 0.2 \text{tc}, \quad \sigma(\mathbf{x}) = 100 \text{tc}^{-1} + 5 \text{AR} + 20(1 - \text{TR})$$

Agentic AI in MDO (4/4): Illustrative Example - Wing MDO



LLM Engagement: Orchestration Layer

Planner Agent

- Orchestrates phases: sampling, optimization, and review
- Issues goals to numerical agents (CFD/FEM/surrogate)
- Reports progress and explains next-step rationale

Reviewer Agent

- Interprets feasible candidate designs
- Ranks alternatives by objective value and risk
- Produces concise engineer-facing recommendations

Division of tasks

The LLM handles strategy, coordination, and communication; numerical solvers and optimizers handle heavy computation.

Bayesian Deep-Ensemble — Architecture

Network structure (per member)

- Two shared hidden layers with ReLU activations
- **Dual output heads:**
 - $\hat{\mu}(\mathbf{x})$ — predicted mean
 - $\hat{\sigma}^2(\mathbf{x}) = e^{\widehat{\log \sigma^2(\mathbf{x})}}$ — predicted variance
- Trained by minimising the heteroscedastic NLL loss:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \frac{1}{2} \left[\log \hat{\sigma}^2(\mathbf{x}_i) + \frac{(y_i - \hat{\mu}(\mathbf{x}_i))^2}{\hat{\sigma}^2(\mathbf{x}_i)} \right]$$

Ensemble aggregation (M members)

Mixture mean:

$$\bar{\mu} = \frac{1}{M} \sum_{m=1}^M \hat{\mu}_m$$

Mixture variance (law of total variance):

$$\bar{\sigma}^2 = \underbrace{\frac{1}{M} \sum_m \hat{\sigma}_m^2}_{\text{aleatoric}} + \underbrace{\frac{1}{M} \sum_m \hat{\mu}_m^2 - \bar{\mu}^2}_{\text{epistemic}}$$

Models Training & Retraining Schedule

Lifecycle

- 1 Run **initial direct simulations** (default: 20) — no surrogate yet.
- 2 Once the budget is reached, fit the ensemble on all simulated data.
- 3 After every **retrain interval** of new direct simulations (default: 5), refit the ensemble with the enlarged dataset.

Input normalisation

$$\tilde{\mathbf{x}} = \frac{\mathbf{x} - \mu_x}{\sigma_x}$$

Data used for training

- Only *simulated* (true-physics) evaluations enter the training set — predicted evaluations are **never** used to retrain.
- Each ensemble member is seeded differently ($\text{seed} = 1000 + \text{member_index}$) to encourage diversity.

Warm-start

If a prior evaluation CSV exists, the surrogate is bootstrapped from it — no wasted simulations across restarts.

Prediction Pipeline — From Input to Probabilities

① Forward pass through each ensemble member

$$\hat{\mu}_m(\mathbf{x}), \quad \hat{\sigma}_m^2(\mathbf{x}) = e^{\widehat{\log \sigma^2}_m(\mathbf{x})}, \quad m = 1, \dots, M$$

② Aggregate into a single Gaussian predictive distribution $\mathcal{N}(\bar{\mu}, \bar{\sigma}^2)$ using the mixture equations (Slide 24).

③ Monte-Carlo sampling — draw $S = 2000$ samples $\{y^{(s)}\}_{s=1}^S \sim \mathcal{N}(\bar{\mu}, \bar{\sigma}^2)$ for each output (objective + constraint metrics).

④ Estimate probabilities

Probability of infeasibility

$$\hat{P}_{\text{inf}} = \frac{1}{S} \sum_{s=1}^S \mathbf{1}[\exists j : c_j^{(s)} > 0]$$

where $c_j^{(s)}$ is the j -th constraint evaluated on the s -th sample.

Probability of no improvement

$$\hat{P}_{\text{ni}} = \frac{1}{S} \sum_{s=1}^S \mathbf{1}[f^{(s)} \geq f^* - \varepsilon]$$

where f^* is the best feasible objective found so far and ε is the improvement margin.

Prediction Acceptance — When to Trust the AI Model?

Acceptance rule

An AI model prediction is *accepted* (i.e. the true simulator is **skipped**) if and only if:

$$\hat{P}_{\text{inf}} \geq \tau \quad \text{OR} \quad \hat{P}_{\text{ni}} \geq \tau$$

Intuition

- High $\hat{P}_{\text{inf}}$: the AI model is confident the point is infeasible — no need to waste a simulation on it.
- High $\hat{P}_{\text{ni}}$: the AI model is confident the point won't improve the best known objective — safe to skip

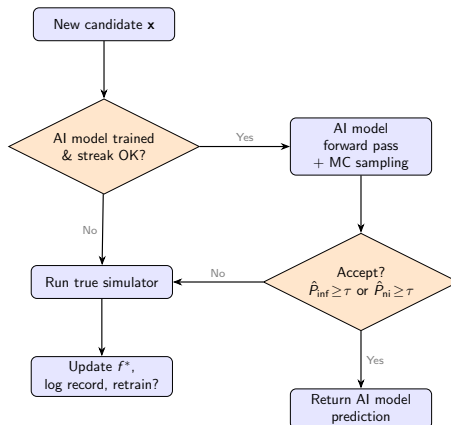
Threshold levels τ

Level	τ	Character
conservative	0.85	Cautious
average	0.65	Balanced
predictive	0.50	Aggressive

If prediction is rejected:

- 1 Run the true simulator.
- 2 Update f^* if feasible & improving.
- 3 Log $\hat{P}_{\text{inf}}, \hat{P}_{\text{ni}}$ alongside the true result for post-analysis.
- 4 Trigger retraining if the interval is reached.

Overall Evaluation Flow



Simulated evaluations

Always logged; used for training and f^* tracking.

Predicted evaluations

Logged with $\hat{P}_{inf}$ and $\hat{P}_{ni}$; **never** used to retrain the surrogate.

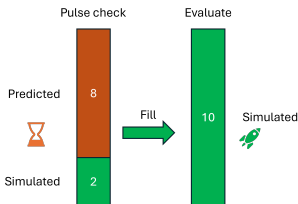
Concurrency in Parallel MDO: Tree of Parzen Estimators — Core Idea

Setup

Given a set of evaluated candidates, split them into two groups using a quantile threshold γ :

$$\mathbf{x} \in \begin{cases} \ell(\mathbf{x}) & \text{if } f(\mathbf{x}) < f^{(\gamma)} & \text{(simulated)} \\ g(\mathbf{x}) & \text{if } f(\mathbf{x}) \geq f^{(\gamma)} & \text{(predicted)} \end{cases}$$

where $f^{(\gamma)}$ is the γ -quantile of observed values (typically $\gamma = 0.15$).

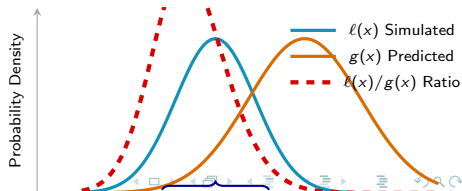


Instead of modelling $p(f | \mathbf{x})$ directly, TPE models two **non-parametric densities**:

- $\ell(\mathbf{x})$ — KDE over *simulated* (low-objective) points
- $g(\mathbf{x})$ — KDE over *predicted* (high-objective) points

Expected Improvement is then proportional to:

$$\text{EI}(\mathbf{x}) \propto \frac{\ell(\mathbf{x})}{g(\mathbf{x})}$$



Methodology applied to Wing Design Problems

AI model targets and uncertainty:

$$\mathbf{y}(\mathbf{x}) = [C_d, C_l, \sigma]^\top, \quad \hat{\mathbf{y}}^{(k)} \sim \mathcal{N}(\boldsymbol{\mu}^{(k)}(\mathbf{x}), \text{diag}(\boldsymbol{\sigma}^{2,(k)}(\mathbf{x})))$$

$$\bar{\boldsymbol{\mu}}, \bar{\boldsymbol{\sigma}}^2 \quad \text{from ensemble moment matching}$$

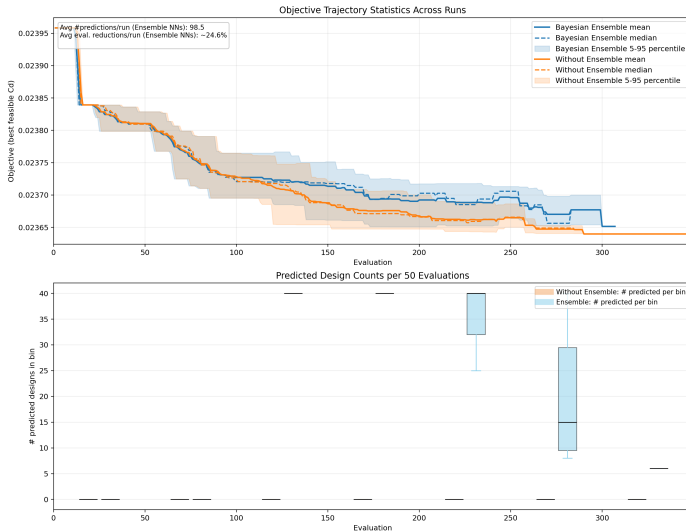
Risk metrics used in logging/decision support:

$$\hat{p}_{\text{inf}}(\mathbf{x}) = \mathbb{P}(|C_l - 1.2| > 0.1 \vee \sigma > 800)$$

$$\hat{p}_{\text{no-imp}}(\mathbf{x}) = \mathbb{P}(C_d \geq C_d^* - \delta), \quad C_d^* = \min_{\text{feasible sampled } i} C_{d,i}$$

Optimization loop (OMADS + AI model): Initial true evaluations $\rightarrow$ train AI model $\rightarrow$ alternate predicted/true evaluations with capped predicted streak and periodic retraining.

Results: Wing Design Validation

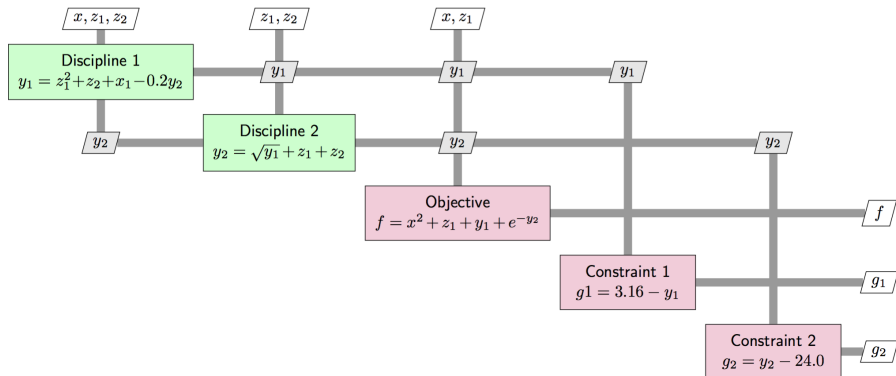


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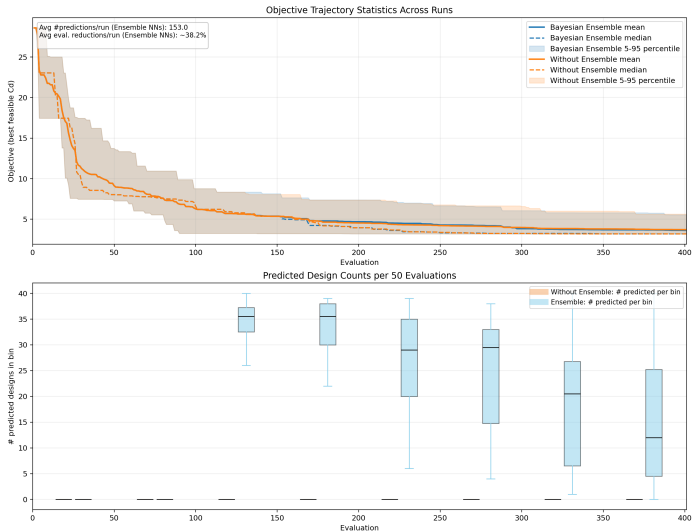
Sellar Benchmark Problem (1/2)

Purpose: compact benchmark for coupled-discipline MDO.



XDSM: optimizer with two coupled Sellar analyses and feedback exchange.

Sellar Benchmark Problem (2/2)

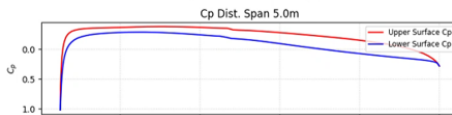
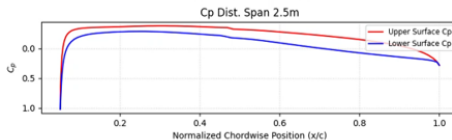
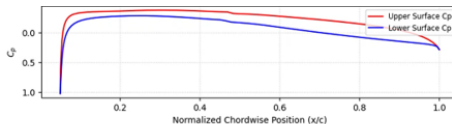
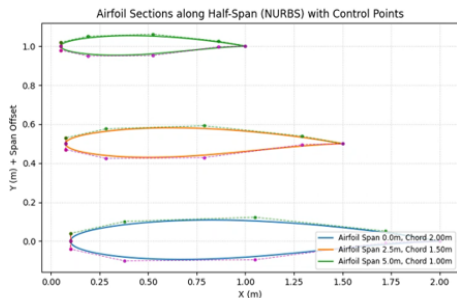


Wing Airfoil Optimization - XFOIL (1/3)

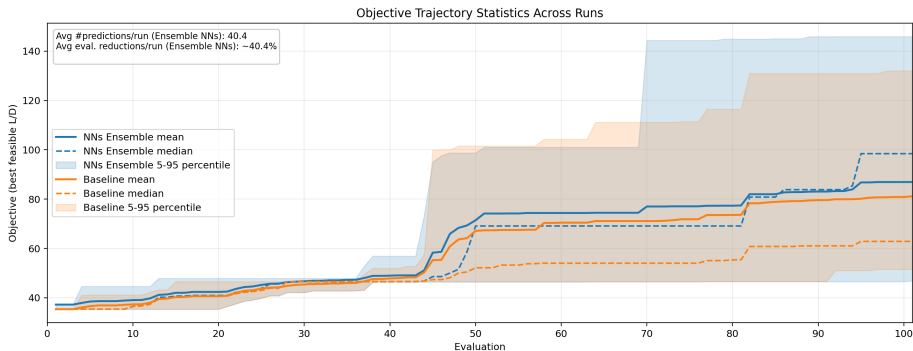
$$\begin{aligned} \min_{\mathbf{x}, \mathbf{u}_a, \mathbf{u}_s} \quad & f(\mathbf{x}) = -C_L(\mathbf{x})/C_D(\mathbf{x}, \mathbf{u}_a) \\ \text{s.t.} \quad & g_1(\mathbf{x}, \mathbf{u}_s) = \sigma_{\max}^{\text{wing}}(\mathbf{x}, \mathbf{u}_s) - \sigma_{\text{allow}} \leq 0, \\ & g_2(\mathbf{x}, \mathbf{u}_s) = C_{D_{\text{allow}}}(\mathbf{x}, \mathbf{u}_s) - C_{D_{\text{allow}}} \leq 0, \end{aligned}$$

$\mathbf{x}$: NURBS control-point perturbations; $\mathbf{u}_a$: aerodynamic state (XFOIL); $\mathbf{u}_s$: structural state (beam model).

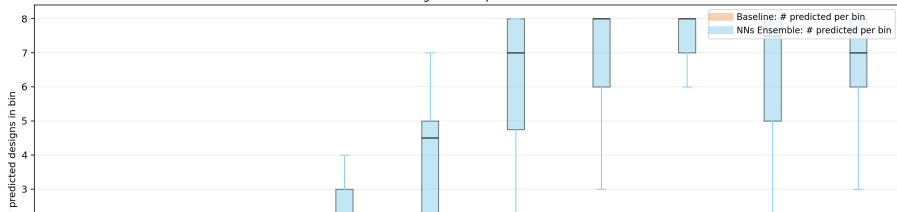
Wing Airfoil Optimization - XFOIL (2/3)



Wing Airfoil Optimization - XFOIL (3/3)



Predicted Design Counts per 10 Evaluations



Supersonic Business Jet Problem

SBJ benchmark (DMDO/NHATC): A distributed multidisciplinary optimization problem for a supersonic business jet with strong couplings and nonlinear constraints.

 [Open in Google Colab](#)

- **Objective:** minimize aircraft weight while satisfying mission/performance requirements.
- **Disciplines:** aircraft performance, propulsion, aerodynamics, and structures.
- **Why it matters:** a compact but realistic benchmark for coordination under coupled constraints.

Context from Notebook

The formulation coordinates subsystem targets using NHATC in a distributed MDO setting.

Outline

- 1 Introduction
 - Multidisciplinary analysis (MDA)
 - Monolithic multidisciplinary design optimization (MDO)
 - Distributed MDO
 - MDO Challenges
- 2 AI/ML in MDO
 - AI Approaches in MDO
 - Bayesian NNs Deep-Ensemble
 - Concurrency voids in Parallel MDO
 - Illustrative Example — Wing MDO
- 3 Case Studies
 - Case Study 1 — Sellar Benchmark Problem
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Machine learning models

- Neural Networks (BNN, FFNN, RNN)
- Kernel methods (GP, RBF)
- Tree-based models (Random Forest, XGBoost)
- Ensemble methods (stacking, boosting)

Derivative-Free Optimization

- Direct search (MADS, GPS)
- Evolutionary algorithms (GA)
- Heuristic methods (SA, PSO)
- Bayesian Optimization

Gradient-Based (Differentiable)

- Adjoint methods
- Physics-Informed Neural Nets
- Differentiable Simulators

Reinforcement Learning

- Bootstrapping
- Uncertainty-aware exploration
- Multi-agent discipline coordination

MDO Frameworks

- **DMDO** — McGill (NHATC adopted by Siemens), open-source
- **OpenMDAO** — NASA
- **HEEDSConnect** — Siemens DISW, Commercial

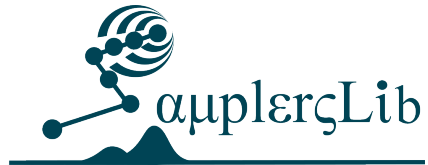
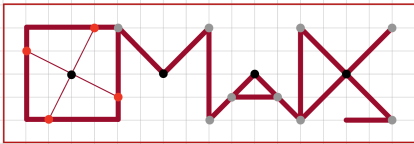
Optimization

- OMADS, Pareto MO, SO
- NSGA-II
- Ax / BoTorch (Bayesian)

Surrogate & ML

- scikit-learn (GP, RBF)
- PyTorch / TensorFlow
- SMT — Surrogate Modeling Toolbox
- DeepXDE (PINNs)

Tools for today's exercises



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Workshop Exercises Overview

#	Exercise	Tools
1	LLM in MDO	scikit-learn
2	Sellar MDO	pytorch+OMADS
3	Airfoil Optimization	XFOIL+pytorch+OMADS
4	Coupled SBJ MDO	DMD0+pytorch+OMADS

Setup

All notebooks available at:

<https://github.com/Ahmed-Bayoumy/AIAA-WS2026-AIMDO>

```
pip install omads dmdo samplersLib pytorch deepxde
```

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Where Is the Field Heading?

Near-term (1–3 years)

- Foundation models for simulation (AeroGPT, ClimateNet)
- LLM-assisted problem formulation
- Uncertainty quantification as first-class citizen
- AI-driven collaborative MDO platforms

Long-term (5–10 years)

- Fully autonomous design agents
- Real-time MDO in digital twins
- Human-in-the-loop preference learning
- Quantum-enhanced optimization

Open Challenges

- ❶ **Scalability** — curse of dimensionality for $d > 100$ design variables
- ❷ **Extrapolation** — surrogates/Al fail outside the training region
- ❸ **Certification** — regulators need explainable, verifiable AI-driven designs
- ❹ **Data scarcity** — high-fidelity runs are expensive; transfer learning is key
- ❺ **Multi-objective** — Pareto front approximation in AI-driven MDO settings
- ❻ **Human factors** — bridging the gap between AI recommendations and designer intuition

Thank You!

Questions & Discussion



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`github.com/Ahmed-Bayoumy/AIAA-WS2026-AIMDO`



`https://ahmed-bayoumy.github.io/`